

Emon Dey

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RESEARCH INTERESTS

High Performance Computing, Federated Learning, Deep Model Compression, Robotics, Generative AI

EDUCATION

University of Maryland, Baltimore County (UMBC)

August 2019 – April 2025

Ph.D. in Information Systems, AI/ML Track

Dissertation Title: A Robust and Resilient Federated Learning Framework over Lossy Communication Channels

Advisor: [Dr. Nirmalya Roy](#)

M.S. in Information Systems, AI/ML Track, May 2022

CGPA: 3.97/4.0

WORK EXPERIENCE

Postdoctoral Appointee, MCS Division, Argonne National Laboratory

April 2025 – Present

Advisor: [Dr. Kibaek Kim](#)

Lemont, IL

- Multi-modal Vision Transformer for phase retrieval in diffraction imaging.
- Queue-aware asynchronous federated learning for high-performance computing applications.
- Federated learning for electrical smart grid systems.
- Continual federated learning.

Graduate Research Assistant, CARDS Lab, UMBC

August 2021 – April 2025

Research Assistant

Baltimore, MD

- Conducted research on decentralized multi-master ROS systems for communication between UAVs and UGVs.

Graduate Teaching Assistant, Department of Information Systems, UMBC

August 2020 – May 2021

Teaching Assistant

Baltimore, MD

- Supported courses on Decision Support Systems, Machine Learning Basics, and Database Application Development.

Graduate Research Assistant, MPSC Lab, UMBC

August 2019 – July 2020

Research Assistant

Baltimore, MD

- Worked on accelerating transfer learning on edge devices by reducing negative transfer.
- Developed DeepIoT, a framework to recognize human activity on IoT devices using deep models.

Lecturer, Port City International University

September 2018 – July 2019

Lecturer

Bangladesh

- Taught Digital Signal Processing, Professional Ethics, Power Systems, and Electrical Machine Design.

Student Mentor, NSF-REU, UMBC

Summer 2021, 2022, 2023

Research Mentor

Baltimore, MD

- Supervised project on synchronizing middleware for heterogeneous multi-robot systems with low-latency co-simulation.
- Supervised project on federated learning for camera-based contactless heart-rate monitoring.
- Supervised project on resource-efficient quantized federated learning for brain-tumor segmentation.
- Supervised project on addressing statistical heterogeneity in federated learning for sea-ship datasets.

TECHNICAL STRENGTHS

Programming Languages

C, C++, MATLAB, Python, R

High Performance Computing

MPI, OpenMP, PyTorch Distributed

Simulation Software

Gazebo, MAVROS, PX4, Unity3D, NS-3, WEKA, Tableau

Deep Learning Frameworks

Keras, PyTorch, TensorFlow

Specialized Tools

OpenCV, TFF, PySyft, FedML, APPFL, TF-Lite, ONNX, Q-Keras, ROS

Hardware Proficiency

Raspberry Pi, Jetson Nano, Google Coral Dev Board, Intel NCS2, Arduino

Soft Skills

Leadership, Communication, Goal-Oriented, Problem Solving, Adaptability

Others

Shell Scripting, Linux, Docker, Git

CURRENT PROJECTS

Project 1: SYNAPS-I: Synergistic Neutron and Photon Autonomous Science-Imaging October 2025 – Ongoing

- Designed a physics-aware Vision Transformer for 2D ptychography generalized across thousands of probes.

- Optimized the ViT model for large-scale training on Perlmutter and Aurora, scaling up to 24,576 GPUs.
- Designed a fine-tuning pipeline with the trained ViT model to facilitate real-time deployment at the beamline.

Project 2: GridMind: Powering the Control Room of the Future with AI Agents March 2026 – Ongoing

- Investigating data and model scaling performance of the proposed Lumina model with heterogeneous graph architecture.
- Designing IsoFLOP experiments to reveal compute-optimal balance between model size and training data allocation.
- Deploying the Lumina model across multiple DOE supercomputers to conduct federated training.

Project 3: AI4S: Continual Federated Learning under Intermittent Client Participation June 2025 – Ongoing

- Collaborating to develop the foundations of ensemble continual learning in federated learning settings.
- Explored the impact of arbitrary arrival and dropping-off of client nodes on server aggregation.
- Designed the experimental setting to test the proposed ensemble continual learning algorithm.

Project 4: AI4S: Asynchronous Federated Learning for High Performance Computing August 2025 – Ongoing

- Tackled varied queuing time across multiple DOE supercomputers while deploying federated learning.
- Implemented a queue-delay-aware asynchronous training framework and tested it in simulation.
- Successfully deployed the proposed framework across Aurora, Perlmutter, Polaris, and Frontier.

COMPLETED PROJECTS

Project 1: Remote Robotic Experimentation using Distributed Virtual Proving Ground (DVPG), DEVCOM Army Research Lab Summer 2021 – Spring 2025

- Demonstrated components that leverage ARL’s Distributed Virtual Proving Ground infrastructure.
- Conducted field experimentation to demonstrate class-incremental federated learning.
- Deployed a lightweight simultaneous localization and mapping system on Clearpath Jackal and SPOT.
- Monitored robotic agents’ movements and actions from a long-distance remote location.

Project 2: Robust Deep Learning Models for Securing Multi-domain Autonomous Cyber-Physical Systems, Office of Naval Research (ONR) Spring 2023 – Spring 2025

- Developed a model-poisoning-attack-resistant federated learning approach for camouflaged and occluded object detection.
- Developed a robust federated learning approach for camouflaged and occluded multi-class object detection.
- Integrated adaptive quantization-based compressive modeling to reduce communication overhead.
- Tackled global and local class imbalances using customized Layer-wise Relevance Propagation.

Project 3: Resilient and Effective Communication and Sensing for Autonomous Platforms and Systems, National Science Foundation (NSF) Spring 2021 – Spring 2025

- Explored communication challenges in non-line-of-sight scenarios within ROS-compatible networks.
- Designed data-security-aware low-power communication protocols in adversarial contexts.
- Proposed a modified sliding-window protocol for high velocity disparity among robotic agents.

Project 4: Asynchronous Federated Training in Multi-Modal Contested Environment, DEVCOM Army Research Lab Fall 2022 – Spring 2024

- Developed a multi-modal deep model for object detection using image and acoustic data in contested environments.
- Addressed data distribution, synchronization, communication overhead, and security issues for asynchronous federated learning.
- Proposed a multi-modal federated training setup to reduce resource overhead using 2-bit quantization-based compression.

PRESENTATIONS AND TALKS

- April 15, 2025, “Quantized Federated Learning over Data Imbalance and Lossy Communication Channels,” IS 698/800: Selected Topics in Machine Learning: Advanced Deep Learning Techniques.
- October 17, 2024, “Introduction to Crazyflie Functionalities and ROS Integration,” IS 709/809: Computational Methods for IS Research.
- September 17, 2024, “Towards Resilient and Secure Federated Learning: Tackling Non-IID Data, Communication Challenges, and Data Security Threats,” Candidate Seminar, Mathematics and Computer Science Division, Argonne National Laboratory.
- February 21–22, 2024, “Resilient and Effective Communication and Sensing for Autonomous Vehicles and Systems,” ArtIAMAS Winter Meeting and Annual Review, ARL Adelphi Laboratory Center.
- February 13–14, 2023, “A Reliable and Low-latency Synchronizing Middleware for Co-simulation of a Heterogeneous Multi-robot System,” ArtIAMAS Year 3 Kickoff Meeting, University of Maryland, Baltimore County campus.
- September 29, 2022, “Robot Operating System (ROS),” IS 709/809: Computational Methods for IS Research.

- March 16–17, 2022, “Decentralized Multi-master ROS System for Communication Between UGVs and UAVs,” ArtIAMAS Annual Review, University of Maryland, College Park campus.
- November 3, 2020, “Neural Networks,” IS 425: Decision Support Systems.

PEER-REVIEWED PUBLICATIONS

J: Journal, C: Conference, W: Workshop.

Journal Articles

- [J2] Anwar, M. S., Ravi, A., **Dey, E.**, Ghosh, I., Ahmed, M., and Roy, N. “MobHeteroCAS: Mobility-Aware DNN Task Scheduling in Heterogeneous Collaborative Autonomous Systems.” *ACM Transactions on Autonomous and Adaptive Systems*, 2025.
- [J1] Shinde, G., Ravi, A., **Dey, E.**, Sakib, S., Rampure, M., and Roy, N. “A Survey on Efficient Vision-Language Models.” *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 15(3), e70036, 2025.

Conference Papers

- [C18] Li, Y., **Dey, E.**, Li, Z., Raghavan, K., Madduri, R., and Kim, K. “FedQueue: Queue-Aware Federated Learning for Cross-Facility HPC Training.” *International Conference on Machine Learning, ICML 2026*.
- [C17] **Dey, E.**, Ravi, A., Shinde, G., Chugh, G., Ghosh, I., Misra, A., and Roy, N. “Fed-CASQ: Enhancing Class-Wise Accuracy in Pervasive Federated Learning with Class-Aware Scaling and Quantization.” *IEEE International Conference on Pervasive Computing and Communications, PerCom 2026*.
- [C16] **Dey, E.**, Hasan, Z., and Roy, N. “FedGALP: Federated Generative Adversarial Learning for Remote Photoplethysmography.” *IEEE Computer Society Annual Symposium on VLSI, ISVLSI 2026*.
- [C15] Hossain, J., **Dey, E.**, Chugh, S., Ahmed, M., Anwar, M. S., Faridee, A. Z., and Roy, N. “SERN: Simulation-Enhanced Realistic Navigation for Multi-Agent Robotic Systems in Contested Environments.” *IEEE International Conference on Computer Communications and Networks, ICCCN 2026*.
- [C14] Shinde, G., Ravi, A., **Dey, E.**, Lewis, J., and Roy, N. “TAVIC-DAS: Task and Channel-Aware Variable-Rate Image Compression for Distributed Autonomous System.” *IEEE Workshop on Pervasive and Resource-constrained Artificial Intelligence, PerConAI 2025*.
- [C13] Sakib, S., Shinde, G., **Dey, E.**, and Roy, N. “E2RespUNet: End-to-End Respiratory Signal Reconstruction and Rate Prediction Using a Unified Attention-Enhanced U-Net.” *International Conference on Smart Computing, SMART-COMP 2025*.
- [C12] Anwar, M. S., Ravi, A., **Dey, E.**, Ghosh, I., Ahmed, M., and Roy, N. “CoOpTex: Multimodal Cooperative Perception and Task Execution in Time-critical Distributed Autonomous Systems.” *International Conference on Distributed Computing in Smart Systems and the Internet of Things, DCOSS-IoT 2025*.
- [C11] **Dey, E.**, Ravi, A., Lewis, J., Kumar, V. K., Freeman, J., Gregory, T., Suri, N., Busart, C., and Roy, N. “DACC-Comm: DNN-Powered Adaptive Compression and Flow Control for Robust Communication in Network-Constrained Environments.” *International Conference on Communication Systems and Networks, COMSNETS 2025*.
- [C10] **Dey, E.**, Walczak, M., Anwar, M. S., Roy, N., Freeman, J., Gregory, T., and Busart, C. “A Novel ROS2 QoS Policy-Enabled Synchronizing Middleware for Co-Simulation of Heterogeneous Multi-Robot Systems.” *IEEE International Conference on Computer Communications and Networks, ICCCN 2023*.
- [C9] Anwar, M. S., **Dey, E.**, Devnath, M. K., Ghosh, I., Khan, N., Freeman, J., Gregory, T., Suri, N., Jayaraja, K., Ramamurthy, S. R., and Roy, N. “HeteroEdge: Addressing Asymmetry in Heterogeneous Collaborative Autonomous Systems.” *IEEE International Conference on Mobile Ad-Hoc and Smart Systems, MASS 2023*.
- [C8] Devnath, M. K., Chakma, A., Conn, M., Hasan, Z., **Dey, E.**, Anwar, M. S., Pal, B., and Roy, N. “A Systematic Study on Object Recognition Using Millimeter-wave Radar.” *IEEE International Conference on Smart Computing, SMARTCOMP 2023*.
- [C7] Ovi, P. R., **Dey, E.**, Roy, N., Gangopadhyay, A., and Erbacher, R. F. “Towards Developing a Data Security Aware Federated Training Framework in Multi-modal Contested Environments.” *Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications IV, SPIE 2022. Best Paper Award*.
- [C6] **Dey, E.**, Hossain, J., Roy, N., and Busart, C. “SynchroSim: An Integrated Co-simulation Middleware for Heterogeneous Multi-robot System.” *International Conference on Distributed Computing in Sensor Systems, DCOSS 2022*.
- [C5] Hasan, Z., **Dey, E.**, Ramamurthy, S. R., Roy, N., and Misra, A. “RhythmEdge: Enabling Contactless Heart Rate Estimation on the Edge.” *IEEE International Conference on Smart Computing, SMARTCOMP 2022. Best Paper Award*.

- [C4] Ovi, P. R., **Dey, E.**, Roy, N., and Gangopadhyay, A. “ARIS: A Real-Time Edge Computed Accident Risk Inference System.” *IEEE International Conference on Smart Computing*, SMARTCOMP 2021.
- [C3] **Dey, E.**, and Roy, N. “OMAD: On-device Mental Anomaly Detection for Substance and Non-Substance Users.” *IEEE International Conference on Bioinformatics and Bioengineering*, BIBE 2020.
- [C2] Mazumder, A. N., **Dey, E.**, and Majumder, S. “A Color Image Watermarking Scheme Employing the Features of Directive Contrast in the DWT-SVD Domain.” *IEEE International Conference for Convergence in Technology*, I2CT 2019.
- [C1] **Dey, E.**, Majumder, S., and Mazumder, A. N. “A New Approach to Color Image Watermarking Based on Joint DWT-SVD Domain in YIQ Color Space.” *International Conference on Electrical Information and Communication Technology*, EICT 2017.

Workshop Papers and Demonstrations

- [W2] Ovi, P. R., **Dey, E.**, Roy, N., and Gangopadhyay, A. “Mixed Quantization Enabled Federated Learning to Tackle Gradient Inversion Attacks.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, CVPR Workshops 2023.
- [W1] Hasan, Z., **Dey, E.**, Ramamurthy, S. R., Roy, N., and Misra, A. “Demo: RhythmEdge: Enabling Contactless Heart Rate Estimation on the Edge.” *IEEE International Conference on Smart Computing*, SMARTCOMP 2022.

MANUSCRIPT UNDER REVIEW

(C) Raghavan, K., **Dey, E.**, Hahn, S., Munson, T., Li, Y., and Kim, K. (2026). “The Foundations of Ensemble Continual Learning in Federated Learning.” Under review.

RELEVANT GRADUATE COURSES

- Robot Operating Systems
- Statistical Machine Learning
- Deep Learning
- Neuromorphic Computing
- Computational Methods for IS Research
- Smart Home Health Analytics
- Natural Language Processing

PROJECT SHOWCASE

- August 1, 2024, “Cross-institutional Remote Robotics Experimentation using ARL’s Distributed Virtual Proving Ground (DVP) Facility,” ARL Robotics Research Collaboration Campus (R2C2), Graces Quarters, Maryland.
- August 2, 2023, “Enhancing Situational Awareness with Federated Learning using DVP,” ARL Robotics Research Collaboration Campus (R2C2), Graces Quarters, Maryland.
- August 2, 2022, “AI-Enabled Decision Making in Multiple Domains,” ARL Robotics Research Collaboration Campus (R2C2), Graces Quarters, Maryland.
- April 27, 2022, “Deploying Federated Learning via ARL’s DVP Network,” RetriEVER Empowered: Student Success + Research + Community event, University of Maryland, Baltimore County.

AWARDS & PROFESSIONAL ACTIVITIES

- Best Paper Awards at SMARTCOMP 2022 and SPIE 2022.
- Our project on building a foundation model for psychographic reconstruction and real-time beamline deployment featured on APS News, Argonne National Laboratory (2026).
- Our project on developing an agentic AI system as a reasoning co-pilot for power system operators featured on MCS News, Argonne National Laboratory (2026).
- Our project on Remote Robotics Experimentation featured on CBS News, Baltimore (2023).
- Audience Choice Award at UMBC IS Research Symposium.
- Student travel grants for BIBE 2020, SPIE 2022, DCOSS 2022, and ICCCN 2023.
- Reviewer:
Journals: TMC, TIOT, TIST, PMC.
IEEE/ACM Conferences: ICDCN, IE, BDCAT, PerCom, SMARTCOMP, DCOSS-IoT, COMSNETS.